ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jan 2022 P1HR Worked Solutions

Jan 2022 | P1HR

29 questions ▪ question image followed by worked solution

PREPARED BY

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STUDENT EDITION

Worked answers included

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PAST PAPER QUESTION

Topic: **Solving Inequalities**

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Q#: 1 Marks: 3

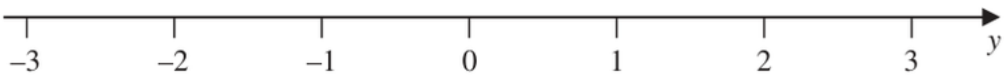
1 n is an integer.

(a) Write down all the values of n such that $-2 \leqslant n < 3$

....

(2)

(b) On the number line, represent the inequality $y \leqslant 1$



(1)

(Total for Question 1 is 3 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 3

TOPIC: **Solving Inequalities**

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WORKED SOLUTION

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Topic check. Solving inequalities. The tag is correct.

Solution

(a)

$$-2 \leq n < 3$$

Since n is an integer,

$$n = -2, -1, 0, 1, 2$$

(b) For

$$y \leq 1$$

use a closed circle at 1 and shade to the left.

FINAL ANSWER

$n = -2, -1, 0, 1, 2$; closed at 1, shaded left.

PAST PAPER QUESTION

Q#: 2 Marks: 2

TOPIC: **Probability Toolkit**

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2 Each time John plays a game, the probability that he wins the game is 0.65

John is going to play the game 300 times.

Work out an estimate for the number of games that John wins.

(Total for Question 2 is 2 marks)

PAST PAPER SOLUTION

Q#: 2 Marks: 2

TOPIC: **Probability Toolkit**

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WORKED SOLUTION

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Topic check. Probability Toolkit. The tag is correct.

Solution

$$0.65 \times 300 = 195$$

FINAL ANSWER

195.

PAST PAPER QUESTION Q#: 3 Marks: 4

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2022 P1HR - Jan 2022 | P1HR

3 The shaded shape is made using three identical right-angled triangles and a square.

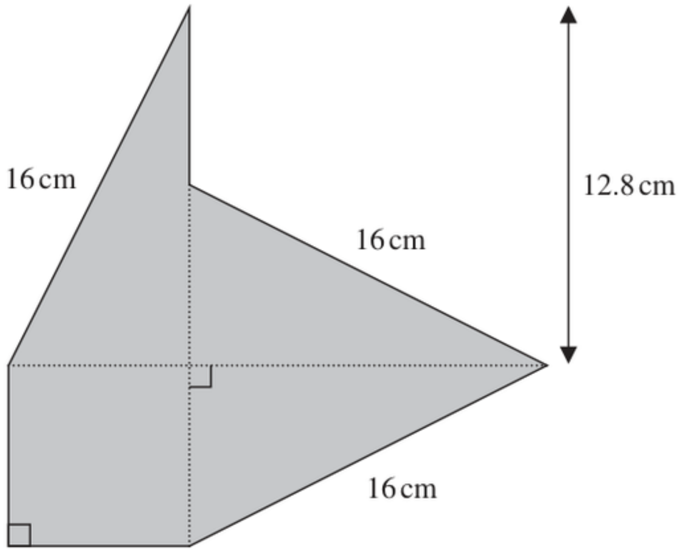

Diagram **NOT** accurately drawn

Work out the perimeter of the shaded shape.

.... cm

(Total for Question 3 is 4 marks)

PAST PAPER SOLUTION Q#: 3 Marks: 4

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION

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Topic check. Right-Angled Triangles - Pythagoras & Trigonometry. The tag is correct.

Solution

Each triangle has hypotenuse 16 cm and one side 12.8 cm.
Find the other side:

$$\sqrt{16^2 - 12.8^2} = 9.6 \text{ cm}$$

The exposed short vertical side is

$$12.8 - 9.6 = 3.2 \text{ cm}$$

Perimeter:

$$3(16) + 9.6 + 9.6 + 3.2 = 70.4$$

FINAL ANSWER
70.4 cm.

PAST PAPER QUESTION

Q#: 4 Marks: 4

TOPIC: **Graphs of Functions**

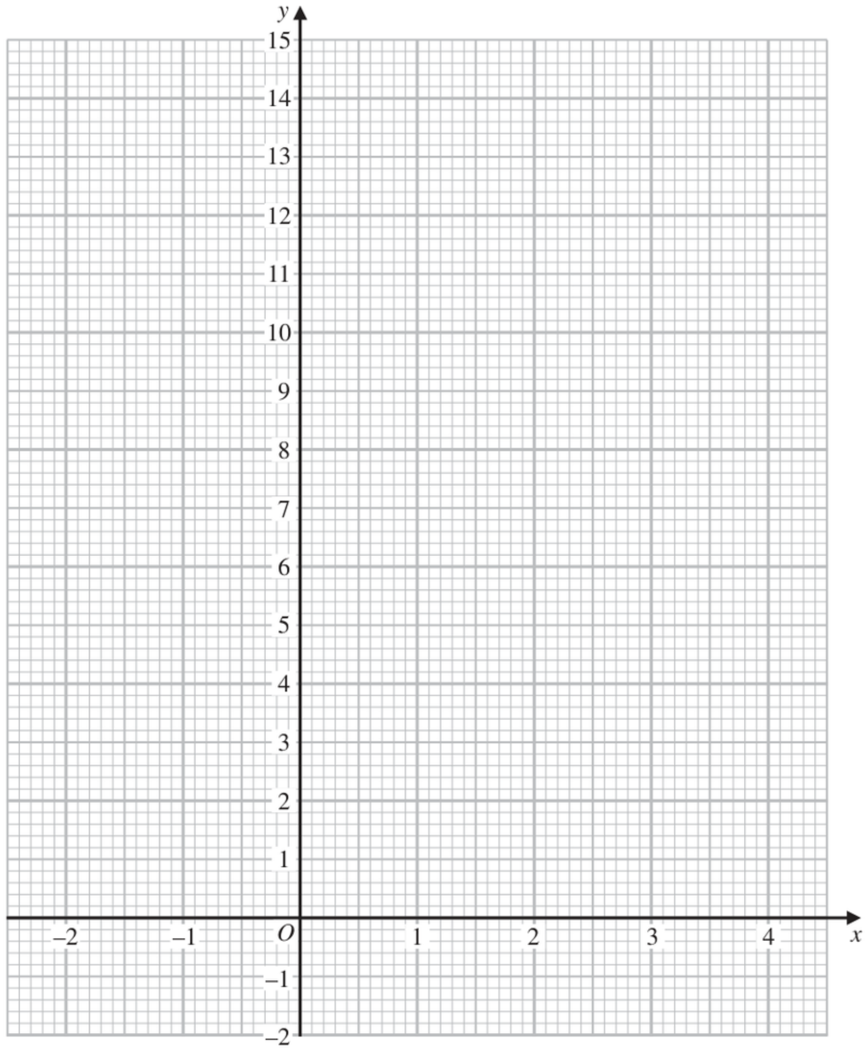
Jan 2022 P1HR - Jan 2022 | P1HR

4 (a) Complete the table of values for $y = x^2 - 4x + 3$

x	-2	-1	0	1	2	3	4
y		8	3			0	

(2)

(b) On the grid, draw the graph of $y = x^2 - 4x + 3$ for values of x from -2 to 4



(2)

(Total for Question 4 is 4 marks)

PAST PAPER SOLUTION

Q#: 4 Marks: 4

TOPIC: **Graphs of Functions**

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Topic check. Correct. This is a table-and-graph question for a quadratic function.

Solution

$$y = x^2 - 4x + 3$$

The completed table is:

x	-2	-1	0	1	2	3	4
y	15	8	3	0	-1	0	3

Plot the points and join them with a smooth U-shaped curve.

FINAL ANSWER

Missing values are $15, 0, -1, 3$.

PAST PAPER QUESTION

Q#: 5 Marks: 4

TOPIC: **Statistics Toolkit**

Jan 2022 P1HR - Jan 2022 | P1HR

5 Yusuf sat 8 examinations.

Here are his marks for 5 of the examinations.

68 72 75 77 80

For his results in all 8 examinations

the mode of his marks is 80
the median of his marks is 74
the range of his marks is 16

Find Yusuf's marks for each of the other 3 examinations.

(Total for Question 5 is 4 marks)

PAST PAPER SOLUTION

Q#: 5 Marks: 4

TOPIC: **Statistics Toolkit**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION

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Topic check. Statistics Toolkit. The tag is correct.

Solution

The range is 16. Since the greatest known mark is 80,

$$80 - 16 = 64$$

So one missing mark is 64.
The mode is 80, so another missing mark is 80.
Now the marks in order are

$$64,\ 68,\ 72,\ x,\ 75,\ 77,\ 80,\ 80$$

The median is 74, so

$$\frac{x+75}{2} = 74$$

$$x = 73$$

FINAL ANSWER

$64,\ 73,\ 80.$

PAST PAPER QUESTION Q#: 6 Marks: 4

TOPIC: **Prime Factors, HCF & LCM**

Jan 2022 P1HR - Jan 2022 | P1HR

6 (a) Work out the lowest common multiple (LCM) of 36 and 120

....
(2)

$A = 5^2 \times 7^4 \times 11^p$
$B = 5^m \times 7^{n-5} \times 11$

m, n and p are integers such that
$m > 2$
$n > 10$
$p > 1$

(b) Find the highest common factor (HCF) of A and B
Give your answer as a product of powers of its prime factors.

....
(2)

(Total for Question 6 is 4 marks)

PAST PAPER SOLUTION

Q#: 6 Marks: 4

TOPIC: **Prime Factors, HCF & LCM**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION

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Topic check. Prime factors, HCF and LCM. The tag is correct.

Method

$$36 = 2^2 \times 3^2$$

$$120 = 2^3 \times 3 \times 5$$

For the LCM, take the highest power of each prime:

$$\text{LCM} = 2^3 \times 3^2 \times 5 = 360$$

For part (b),

$$A = 5^2 \times 7^4 \times 11^p$$

$$B = 5^m \times 7^{n-5} \times 11$$

Since $m > 2$, the smaller power of 5 is 5^2.
Since $n > 10$, $n - 5 > 5$, so the smaller power of 7 is 7^4.
Since $p > 1$, the smaller power of 11 is 11^1.

$$\text{HCF} = 5^2 \times 7^4 \times 11$$

FINAL ANSWER
(a) 360, (b) $5^2 \times 7^4 \times 11$

PAST PAPER QUESTION **Q#: 7** **Marks: 4**

TOPIC: **Standard & Compound Units**

Jan 2022 P1HR - Jan 2022 | P1HR

7 Milly went on a car journey.
She travelled from Anesey to Breigh to Clando and then to Duckbridge.

For Anesey to Breigh, Milly drove the 245 km in 2.5 hours.
For Breigh to Clando, Milly drove the 220 km at an average speed of 80 km/h
For Clando to Duckbridge, Milly drove at an average speed of 72 km/h in 50 minutes.

Work out Milly's average speed, in km/h, for the journey from Anesey to Duckbridge.
Give your answer correct to one decimal place.

.... km/h

(Total for Question 7 is 4 marks)

PAST PAPER SOLUTION

Q#: 7 Marks: 4

TOPIC: **Standard & Compound Units**

Jan 2022 P1HR - Jan 2022 | P1HR

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Topic check. Correct. This is average speed over a whole journey.

Solution

First part:

$$\text{distance} = 245 \text{ km}, \qquad \text{time} = 2.5 \text{ h}$$

Second part:

$$\text{time} = \frac{220}{80} = 2.75 \text{ h}$$

Third part:

$$50 \text{ minutes} = \frac{5}{6} \text{ h}$$

$$\text{distance} = 72 \times \frac{5}{6} = 60 \text{ km}$$

Total distance:

$$245 + 220 + 60 = 525 \text{ km}$$

Total time:

$$2.5 + 2.75 + \frac{5}{6} = 6.0833\ldots \text{ h}$$

Average speed:

$$\frac{525}{6.0833\ldots} = 86.301\ldots$$

FINAL ANSWER

86.3 km/h.

PAST PAPER QUESTION Q#: 8 Marks: 4

TOPIC: **Powers, Roots & Standard Form**

Jan 2022 P1HR - Jan 2022 | P1HR

8 (a) Write 5×10^4 as an ordinary number.

....
(1)

(b) Write 0.000 06 in standard form.

....
(1)

(c) Work out $(4 \times 10^{512}) \div (1.6 \times 10^{700})$
Give your answer in standard form.

....
(2)

(Total for Question 8 is 4 marks)

PAST PAPER SOLUTION

Q#: 8 Marks: 4

TOPIC: **Powers, Roots & Standard Form**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION

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Topic check. Powers, roots and standard form. The tag is correct.

Method

$$5 \times 10^4 = 50000$$

$$0.00006 = 6 \times 10^{-5}$$

For part (c),

$$\frac{4 \times 10^{512}}{1.6 \times 10^{700}} = \frac{4}{1.6} \times 10^{512-700}$$

$$= 2.5 \times 10^{-188}$$

FINAL ANSWER

(a) 50000, (b) 6×10^{-5}, (c) 2.5×10^{-188}

PAST PAPER QUESTION Q#: 9 Marks: 5

TOPIC: **Factorising**

Jan 2022 P1HR - Jan 2022 | P1HR

9 (a) Simplify $x^4 \times x^5$

....

(1)

(b) Simplify $(4y^2)^3$

....

(2)

(c) Factorise $n^2 - 7n + 12$

....

(2)

(Total for Question 9 is 5 marks)

PAST PAPER SOLUTION

Q#: 9 Marks: 5

TOPIC: **Factorising**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION

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Topic check. Factorising. The tag is correct.

Solution

(a) $x^4 \times x^5 = x^9$.
(b) $(4y^2)^3 = 64y^6$.
(c)

$$n^2 - 7n + 12 = (n-3)(n-4)$$

FINAL ANSWER
(a) x^9, (b) $64y^6$, (c) $(n-3)(n-4)$

PAST PAPER QUESTION

Topic: **Volume & Surface Area**

Jan 2022 P1HR - Jan 2022 | P1HR

Q#: 10 **Marks: 6**

10 Jonty has a storage container in the shape of a cuboid, as shown in the diagram.

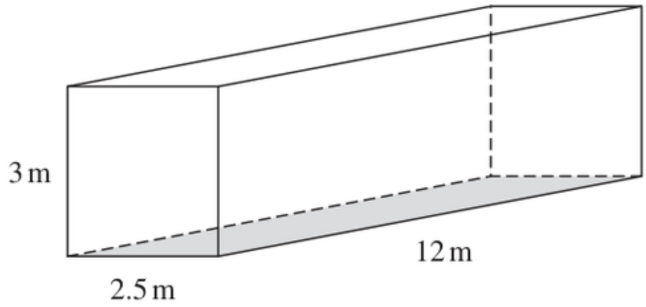

Diagram **NOT** accurately drawn

Jonty is going to paint the outside of his storage container, apart from the base which is shown shaded in the diagram.
He needs enough paint to cover the four sides and the top.

Each tin of paint covers an area of $15\,m^2$

The cost of each tin of paint recently increased by 10%
After the increase, the cost of each tin of paint is £26.95

Jonty says

"**Before** the increase, I could have bought enough paint for less than £200"

Show that Jonty is correct.
Show your working clearly.

(Total for Question 10 is 6 marks)

PAST PAPER SOLUTION

Q#: 10 Marks: 6

TOPIC: **Volume & Surface Area**

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Topic check. Corrected to Volume & Surface Area. The surface area calculation is the main structure of the problem, with a percentage price change at the end.

Method

Jonty paints the four sides and the top, but not the base.
Top:

$$12 \times 2.5 = 30$$

Two long sides:

$$2(12 \times 3) = 72$$

Two short sides:

$$2(2.5 \times 3) = 15$$

Total area:

$$30 + 72 + 15 = 117 \text{ m}^2$$

Each tin covers 15 m^2, so the number of tins needed is

$$\frac{117}{15} = 7.8$$

He must buy 8 tins.
After the 10% increase, each tin costs £26.95. Before the increase:

$$\frac{26.95}{1.10} = 24.50$$

Before the increase, 8 tins would cost

$$8 \times 24.50 = 196$$

Since £196 < £200, Jonty is correct.

FINAL ANSWER
Jonty is correct; before the increase the paint would have cost £196.

PAST PAPER QUESTION

Q#: 11 **Marks: 2**

TOPIC: **Circles, Arcs & Sectors**

Jan 2022 P1HR - Jan 2022 | P1HR

11 The diagram shows sector OPQ of a circle, centre O

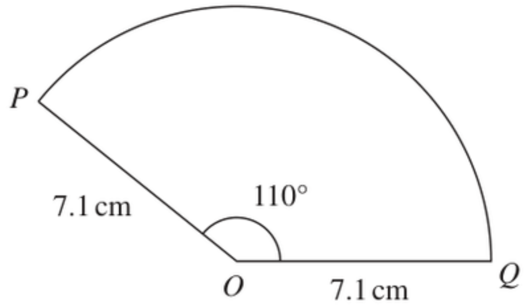

Diagram **NOT** accurately drawn

$OP = OQ = 7.1$ cm
Angle $POQ = 110°$

Calculate the area of sector OPQ
Give your answer correct to one decimal place.

.... cm^2

(Total for Question 11 is 2 marks)

PAST PAPER SOLUTION

Q#: 11 Marks: 2

TOPIC: **Circles, Arcs & Sectors**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION

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Topic check. Circles, Arcs & Sectors. The tag is correct.

Solution

$$\text{area} = \frac{110}{360} \times \pi(7.1)^2$$

$$= 48.390\ldots$$

FINAL ANSWER

$48.4\ \text{cm}^2$.

PAST PAPER QUESTION Q#: 12 Marks: 5

TOPIC: **Algebraic Fractions**

Jan 2022 P1HR - Jan 2022 | P1HR

12 (a) Expand and simplify $n(n-4)(3n+5)$

....
(2)

(b) Express

$$\frac{3}{x} + \frac{x+2}{2x} + \frac{1}{4}$$

as a single fraction in its simplest form.

....
(3)

(Total for Question 12 is 5 marks)

PAST PAPER SOLUTION

Q#: 12 Marks: 5

TOPIC: **Algebraic Fractions**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION

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Topic check. Corrected to algebraic fractions. The original tag was expanding brackets, but the algebraic fraction part is larger.

Solution

(a)

$$n(n-4)(3n+5) = n(3n^2 - 7n - 20) = 3n^3 - 7n^2 - 20n$$

(b)

$$\frac{3}{x} + \frac{x+2}{2x} + \frac{1}{4}$$

Use denominator $4x$:

$$\frac{12}{4x} + \frac{2(x+2)}{4x} + \frac{x}{4x} = \frac{12 + 2x + 4 + x}{4x}$$

$$= \frac{3x+16}{4x}$$

FINAL ANSWER

(a) $3n^3 - 7n^2 - 20n$, (b) $\frac{3x+16}{4x}$

PAST PAPER QUESTION

Q#: 13 Marks: 7

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

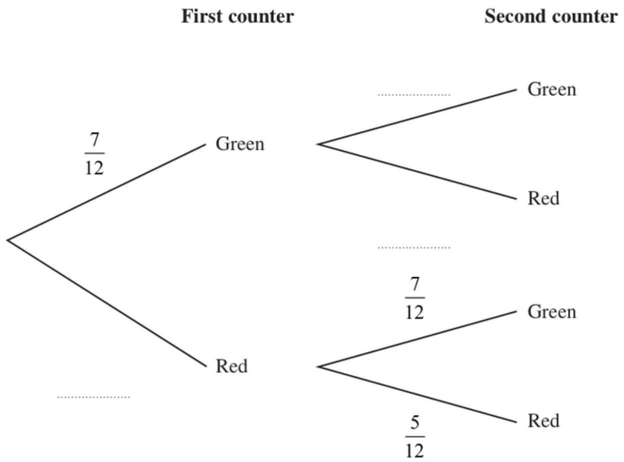
Jan 2022 P1HR - Jan 2022 | P1HR

13 Hector has a bag that contains 12 counters.
There are 7 green counters and 5 red counters in the bag.

Hector takes at random a counter from the bag.
He looks at the counter and puts the counter back into the bag.

Hector then takes at random a second counter from the bag.
He looks at the counter and puts the counter back into the bag.

(a) Complete the probability tree diagram.



(2)

(b) Work out the probability that both counters are red.

....

(2)

Meghan has a jar containing 15 counters.
There are only blue counters, green counters and red counters in the jar.

Hector is going to take at random one of the counters from his bag of 12 counters.
He will look at the counter and put the counter back into the bag.

Hector is then going to take at random a second counter from his bag.
He will look at the counter and put the counter back into the bag.

Meghan is then going to take at random one of the counters from her jar of counters.
She will look at the counter and put the counter back into the jar.

The probability that the 3 counters each have a different colour is $\frac{7}{24}$

(c) Work out how many blue counters there are in the jar.

....

(3)

(Total for Question 13 is 7 marks)

PAST PAPER SOLUTION

Q#: 13 Marks: 7

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Probability Diagrams - Venn & Tree Diagrams. The tag is correct.

Solution

Hector's probability of taking one green and one red is

$$2 \times \frac{7}{12} \times \frac{5}{12} = \frac{35}{72}$$

For all three counters to be different, Meghan must take a blue counter.
Let the number of blue counters in Meghan's jar be b. There are 15 counters in the jar.

$$\frac{35}{72} \times \frac{b}{15} = \frac{7}{24}$$

$$\frac{b}{15} = \frac{3}{5}$$

$$b = 9$$

FINAL ANSWER

9 blue counters.

PAST PAPER QUESTION

Q#: 14 Marks: 4

TOPIC: **Circle Theorems**

Jan 2022 P1HR - Jan 2022 | P1HR

14

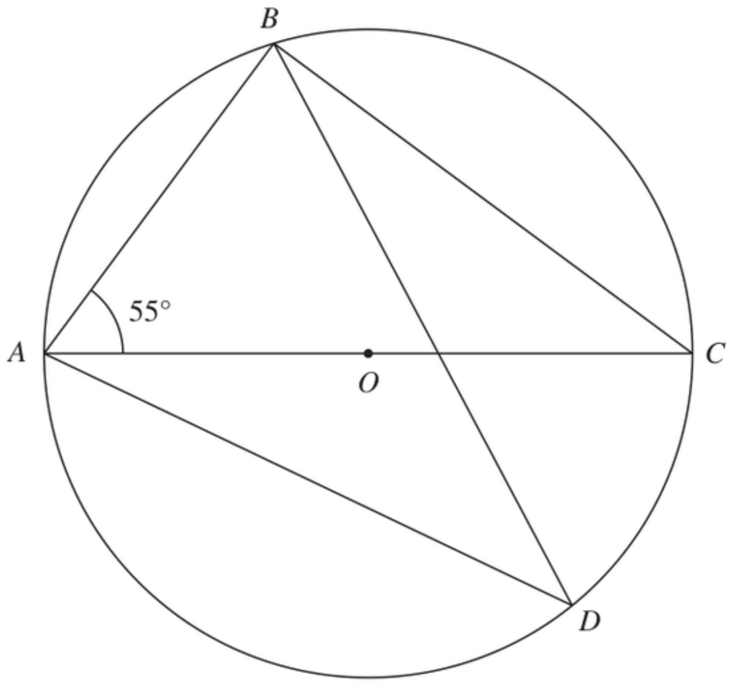


Diagram **NOT** accurately drawn

A, *B*, *C* and *D* are points on a circle, centre *O*
AOC is a diameter of the circle.

Angle *BAC* = 55°

Work out the size of angle *ADB*
Give a reason for each stage of your working.

.... °

(Total for Question 14 is 4 marks)

PAST PAPER SOLUTION

Q#: 14 Marks: 4

TOPIC: **Circle Theorems**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION

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Topic check. Circle Theorems. The tag is correct.

Solution

Since AC is a diameter,

$$\angle ABC = 90^\circ$$

In triangle ABC,

$$\angle ACB = 180^\circ - 90^\circ - 55^\circ = 35^\circ$$

Angles in the same segment are equal, so

$$\angle ADB = \angle ACB = 35^\circ$$

FINAL ANSWER

35°.

PAST PAPER QUESTION

Q#: 15 **Marks: 3**

TOPIC: **Algebraic Proof**

Jan 2022 P1HR - Jan 2022 | P1HR

15 Using algebra, prove that, given any 3 consecutive whole numbers, the sum of the square of the smallest number and the square of the largest number is always 2 more than twice the square of the middle number.

(Total for Question 15 is 3 marks)

PAST PAPER SOLUTION Q#: 15 Marks: 3

TOPIC: **Algebraic Proof**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION

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Topic check. Algebraic proof. The tag is correct.

Solution

Let the three consecutive whole numbers be

$$n, \quad n+1, \quad n+2$$

The sum of the squares of the smallest and largest is

$$n^2 + (n+2)^2 = n^2 + n^2 + 4n + 4 = 2n^2 + 4n + 4$$

Twice the square of the middle number is

$$2(n+1)^2 = 2(n^2 + 2n + 1) = 2n^2 + 4n + 2$$

So

$$2n^2 + 4n + 4 = (2n^2 + 4n + 2) + 2$$

FINAL ANSWER

The sum is always 2 more than twice the square of the middle number.

PAST PAPER QUESTION Q#: 16 Marks: 4

TOPIC: **Sequences**

Jan 2022 P1HR - Jan 2022 | P1HR

16 An arithmetic series has first term 1 and common difference 4

Find the sum of all terms of the series from the 41st term to the 100th term inclusive.

(Total for Question 16 is 4 marks)

PAST PAPER SOLUTION

Q#: 16 Marks: 4

TOPIC: **Sequences**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION

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Topic check. Sequences. The tag is correct.

Solution

The first term is 1 and the common difference is 4.
The 41st term is

$$1 + 40(4) = 161$$

The 100th term is

$$1 + 99(4) = 397$$

There are

$$100 - 41 + 1 = 60$$

terms from the 41st to the 100th inclusive.

$$S = \frac{60}{2}(161 + 397)$$

$$S = 30 \times 558 = 16740$$

FINAL ANSWER

16740.

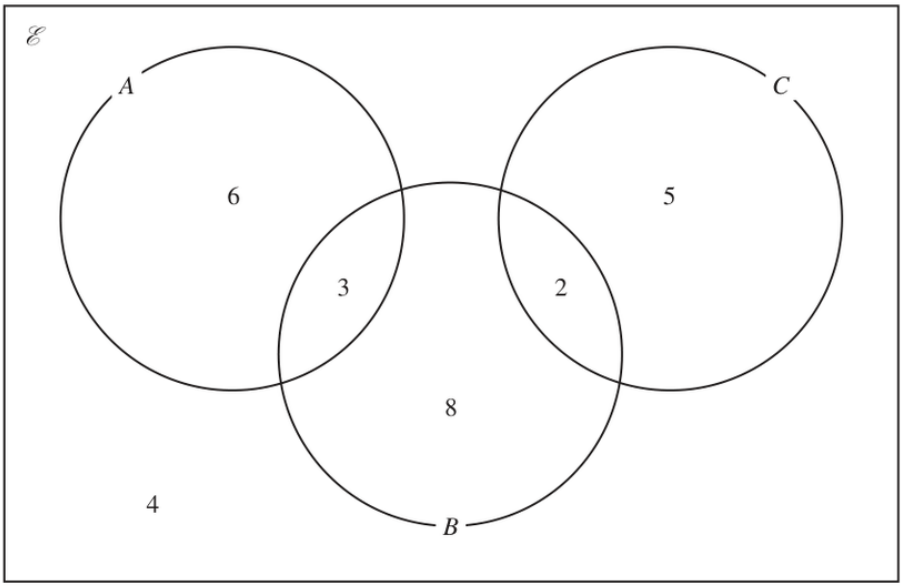
PAST PAPER QUESTION

Q#: 17 Marks: 4

TOPIC: **Set Notation & Venn Diagrams**

Jan 2022 P1HR - Jan 2022 | P1HR

17 The Venn diagram shows a universal set $\mathscr{E}$ and three sets A, B and C.



6, 3, 8, 2, 5 and 4 represent the **numbers** of elements.

Find

(i) $n(A \cup B)$

....
(1)

(ii) $n(A \cap C)$

....
(1)

(iii) $n(B \cap C')$

....
(1)

(iv) $n(A' \cup B' \cup C')$

....
(1)

(Total for Question 17 is 4 marks)

PAST PAPER SOLUTION

Q#: 17 **Marks: 4**

TOPIC: **Set Notation & Venn Diagrams**

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WORKED SOLUTION

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Topic check. Set notation and Venn diagrams. The tag is correct.

Method

From the Venn diagram:

$$A \text{ only} = 6, \quad A \cap B = 3, \quad B \text{ only} = 8, \quad B \cap C = 2, \quad C \text{ only} = 5, \quad \text{outside} = 4$$

There is no region in both A and C, and there is no triple intersection.

$$n(A \cup B) = 6 + 3 + 8 + 2 = 19$$

$$n(A \cap C) = 0$$

$$n(B \cap C') = 3 + 8 = 11$$

For $A' \cup B' \cup C'$, use

$$A' \cup B' \cup C' = (A \cap B \cap C)'$$

Since $A \cap B \cap C$ is empty, this includes everyone in the universal set.

$$6 + 3 + 8 + 2 + 5 + 4 = 28$$

FINAL ANSWER

(i) 19, (ii) 0, (iii) 11, (iv) 28

PAST PAPER QUESTION Q#: 18 Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Jan 2022 P1HR - Jan 2022 | P1HR

18 The three solids **A**, **B** and **C** are similar such that

the surface area of **A** : the surface area of **B** = 4 : 9

and

the volume of **B** : the volume of **C** = 125 : 343

Work out the ratio

the height of **A** : the height of **C**

Give your ratio in its simplest form.

(Total for Question 18 is 4 marks)

PAST PAPER SOLUTION

Q#: 18 Marks: 4

TOPIC: **Area & Volume of Similar Shapes**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION

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Topic check. Corrected to Area and Volume of Similar Shapes. The key idea is converting area and volume ratios into length ratios.

Method

Surface area ratio $A : B = 4 : 9$, so the length ratio is

$$\sqrt{4} : \sqrt{9} = 2 : 3$$

Volume ratio $B : C = 125 : 343$, so the length ratio is

$$\sqrt[3]{125} : \sqrt[3]{343} = 5 : 7$$

Therefore

$$\frac{A}{C} = \frac{2}{3} \times \frac{5}{7} = \frac{10}{21}$$

FINAL ANSWER

10 : 21

PAST PAPER QUESTION

Q#: 19 Marks: 3

TOPIC: **Completing the Square**

Jan 2022 P1HR - Jan 2022 | P1HR

19 Given that $\left(\sqrt[3]{\dfrac{1}{x}}\right)^4 = x^m$

(a) find the value of m

$m =$ ....

(1)

Given that a, b and c are integers,

(b) express $3x^2 + 12x + 19$ in the form $a(x + b)^2 + c$

....

(2)

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 3

TOPIC: **Completing the Square**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION

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Topic check. Completing the square. The tag is correct.

Solution

(a)

$$\sqrt[3]{\frac{1}{x}} = x^{-1/3}$$

So $m = -\frac{1}{3}$.

(b)

$$3x^2 + 12x + 19 = 3(x^2 + 4x) + 19$$

$$= 3\big((x+2)^2 - 4\big) + 19$$

$$= 3(x+2)^2 + 7$$

FINAL ANSWER

$m = -\frac{1}{3}$, and $3(x+2)^2 + 7$

PAST PAPER QUESTION Q#: 20 Marks: 6

TOPIC: **Transformations of Graphs**

Jan 2022 P1HR - Jan 2022 | P1HR

20 The curve with equation $y = f(x)$ has one turning point.

The coordinates of this turning point are $(-6, -4)$

(a) Write down the coordinates of the turning point on the curve with equation

(i) $y = f(x) + 5$

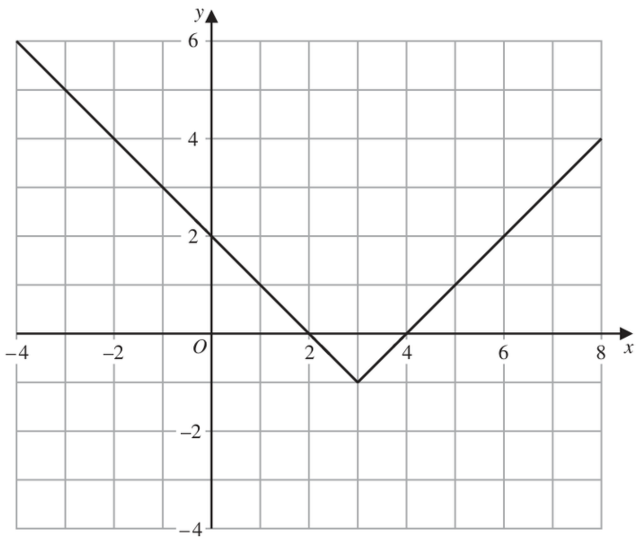
(.................. , ...)

(ii) $y = f(3x)$

(.................. , ...)

(2)

The graph of $y = g(x)$ is shown on the grid below.



(b) On the grid, sketch the graph of $y = 2g(x)$ for $-1 \leqslant x \leqslant 7$

(2)

The graph of $y = h(x)$ intersects the x-axis at two points.
The coordinates of the two points are $(-1, 0)$ and $(6, 0)$

The graph of $y = h(x + a)$ passes through the point with coordinates $(2, 0)$, where a is a constant.

(c) Find the two possible values of a

.... , ...

(2)

(Total for Question 20 is 6 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 6

TOPIC: **Transformations of Graphs**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

The turning point of $y = f(x)$ is $(-6, -4)$.
For

$$y = f(x) + 5$$

the graph moves 5 units up:

$$(-6, -4) \to (-6, 1)$$

For

$$y = f(3x)$$

the x-coordinate is divided by 3:

$$(-6, -4) \to (-2, -4)$$

For the sketch of $y = 2g(x)$, keep every x-coordinate the same and double every y-coordinate.
Useful points are

$$(-1, 3) \to (-1, 6)$$

$$(2, 0) \to (2, 0)$$

$$(3, -1) \to (3, -2)$$

$$(4, 0) \to (4, 0)$$

$$(7, 3) \to (7, 6)$$

Draw straight line segments through these points for $-1 \le x \le 7$.
For part (c), $h(x)$ has roots at $x = -1$ and $x = 6$.
For $h(x + a)$ to pass through $(2, 0)$,

$$2 + a = -1 \quad \text{or} \quad 2 + a = 6$$

$$a = -3 \quad \text{or} \quad a = 4$$

FINAL ANSWER

$(-6, 1)$, $(-2, -4)$, sketch through the doubled points, and $a = -3$ or $a = 4$

PAST PAPER QUESTION

Q#: 20 Marks: 6

TOPIC: **Transformations of Graphs**

Jan 2022 P1HR - Jan 2022 | P1HR

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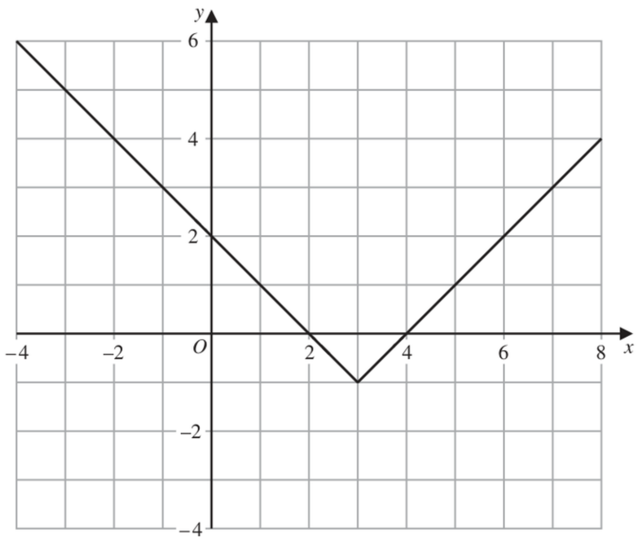

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The graph of $y = h(x + a)$ passes through the point with coordinates $(2, 0)$, where a is a constant.

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(2)

(Total for Question 20 is 6 marks)

PAST PAPER SOLUTION

Q#: 20 Marks: 6

TOPIC: **Transformations of Graphs**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

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FINAL ANSWER

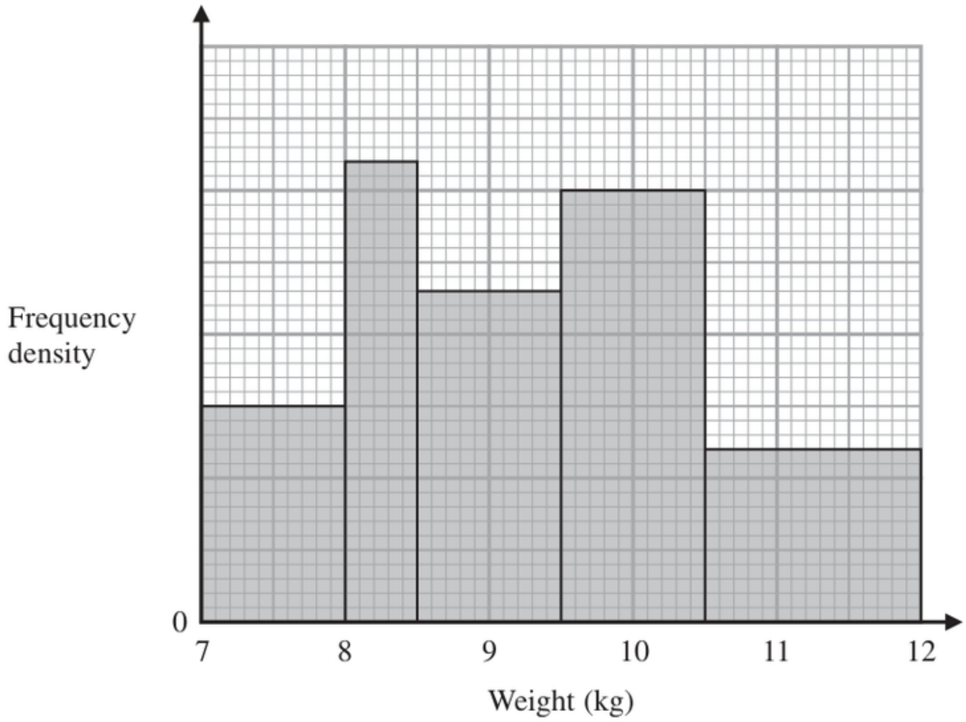
$(-6, 1)$, $(-2, -4)$, sketch through the doubled points, and $a = -3$ or $a = 4$

PAST PAPER QUESTION Q#: 21 Marks: 3

TOPIC: **Histograms**

Jan 2022 P1HR - Jan 2022 | P1HR

21



The histogram gives information about the weights, in kg, of all the watermelons in a field.

There are 16 watermelons with a weight between 8 kg and 8.5 kg

Work out the total number of watermelons in the field.

(Total for Question 21 is 3 marks)

PAST PAPER SOLUTION Q#: 21 Marks: 3

TOPIC: **Histograms**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

For a histogram,

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

For $8 \le w < 8.5$, the frequency is 16 and the class width is 0.5, so

$$\text{frequency density} = \frac{16}{0.5} = 32$$

Using this scale from the histogram, the frequency densities are:

$$15,\ 32,\ 23,\ 30,\ 12$$

The total frequency is

$$(1)(15) + (0.5)(32) + (1)(23) + (1)(30) + (1.5)(12)$$

$$= 15 + 16 + 23 + 30 + 18$$

$$= 102$$

FINAL ANSWER

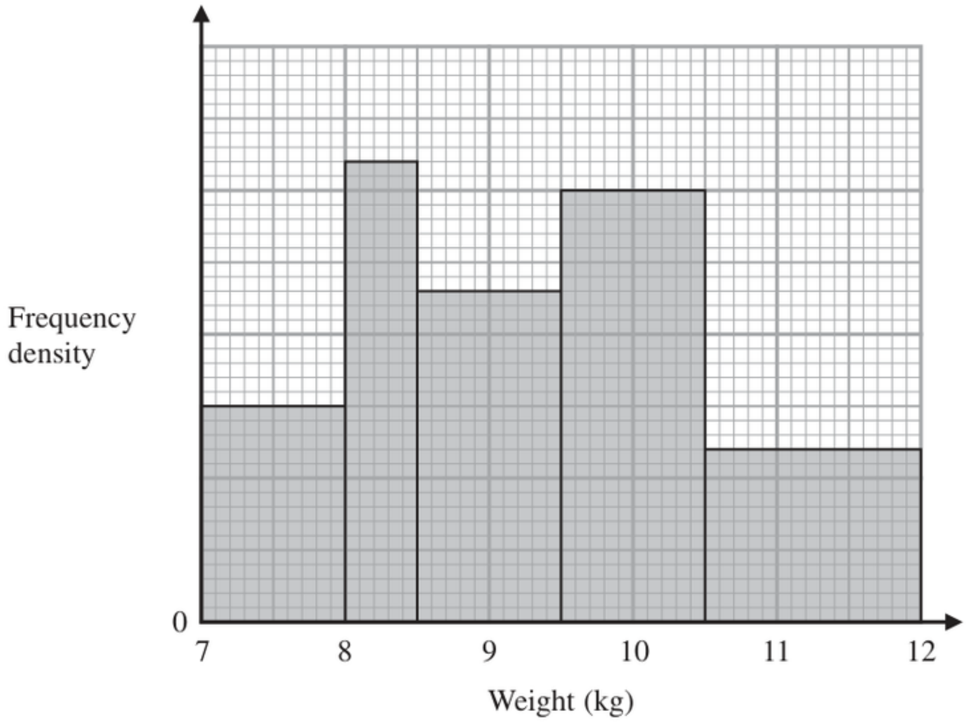
102 watermelons

PAST PAPER QUESTION Q#: 21 Marks: 3

TOPIC: **Histograms**

Jan 2022 P1HR - Jan 2022 | P1HR

21



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PAST PAPER SOLUTION Q#: 21 Marks: 3

TOPIC: **Histograms**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Correct.

Method

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$$= 15 + 16 + 23 + 30 + 18$$

$$= 102$$

FINAL ANSWER

102 watermelons

PAST PAPER QUESTION Q#: 22 Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

Jan 2022 P1HR - Jan 2022 | P1HR

22 The diagram shows triangle *ABC*

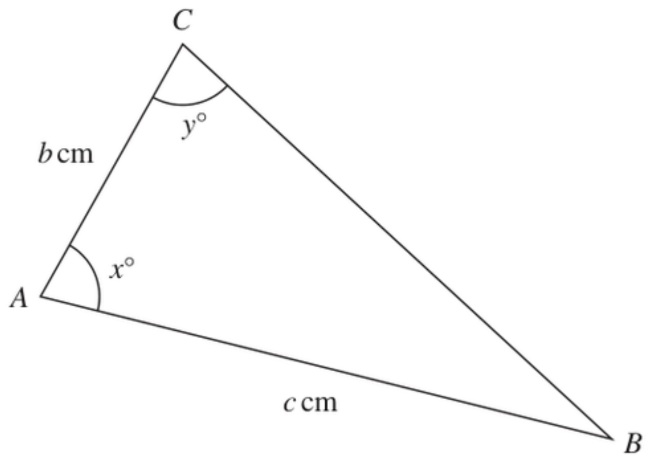


Diagram **NOT** accurately drawn

$c = 11.5$ correct to one decimal place
$x = 80$ correct to the nearest whole number
$y = 75$ correct to the nearest whole number

Calculate the upper bound for the value of b
Show your working clearly.
Give your answer correct to 3 significant figures.

(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Rounding, Estimation & Bounds is correct. The question also uses the sine rule, but the main demand is choosing the correct bounds.

Method

For an upper bound for b, use the upper bound of c:

$c = 11.55$

Use the lower bounds of x and y, because this makes angle B as large as possible and makes $\sin y$ as small as possible.

$x = 79.5°, \quad y = 74.5°$

So

$\angle B = 180° - 79.5° - 74.5° = 26°$

Using the sine rule,

$$\frac{b}{\sin 26°} = \frac{11.55}{\sin 74.5°}$$

$$b = \frac{11.55 \sin 26°}{\sin 74.5°} = 5.254...$$

FINAL ANSWER

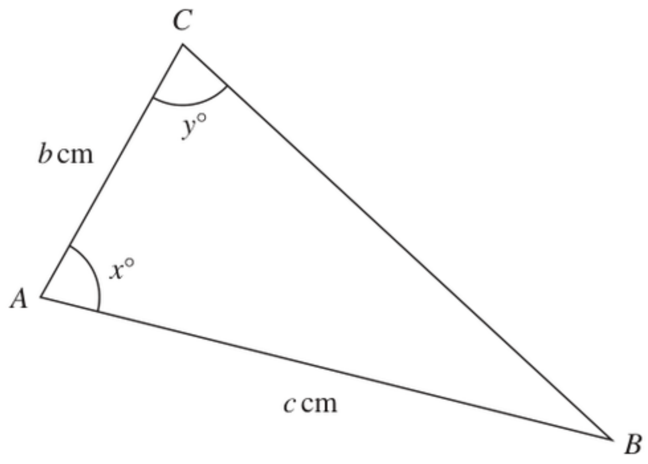
5.25 cm to 3 significant figures.

PAST PAPER QUESTION Q#: 22 Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

Jan 2022 P1HR - Jan 2022 | P1HR

22 The diagram shows triangle ABC



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Calculate the upper bound for the value of b
Show your working clearly.
Give your answer correct to 3 significant figures.

(Total for Question 22 is 4 marks)

PAST PAPER SOLUTION

Q#: 22 Marks: 4

TOPIC: **Rounding, Estimation & Bounds**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

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For an upper bound for b, use the upper bound of c:

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Use the lower bounds of x and y, because this makes angle B as large as possible and makes $\sin y$ as small as possible.

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Using the sine rule,

$$\frac{b}{\sin 26°} = \frac{11.55}{\sin 74.5°}$$

$$b = \frac{11.55 \sin 26°}{\sin 74.5°} = 5.254\ldots$$

FINAL ANSWER

5.25 cm to 3 significant figures.

PAST PAPER QUESTION Q#: 23 Marks: 6

TOPIC: **Differentiation**

Jan 2022 P1HR - Jan 2022 | P1HR

23 Two particles, P and Q, move along a straight line.
The fixed point O lies on this line.

The displacement of P from O at time t seconds is s metres, where

$$s = t^3 - 4t^2 + 5t \quad \text{for } t > 1$$

The displacement of Q from O at time t seconds is x metres, where

$$x = t^2 - 4t + 4 \quad \text{for } t > 1$$

Find the range of values of t where $t > 1$ for which both particles are moving in the same direction along the straight line.

(Total for Question 23 is 6 marks)

PAST PAPER SOLUTION

Q#: 23 Marks: 6

TOPIC: **Differentiation**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Functions to Differentiation.

Method

Particle P:

$$s = t^3 - 4t^2 + 5t$$

$$\frac{ds}{dt} = 3t^2 - 8t + 5$$

$$\frac{ds}{dt} = (3t - 5)(t - 1)$$

Particle Q:

$$x = t^2 - 4t + 4$$

$$\frac{dx}{dt} = 2t - 4 = 2(t - 2)$$

For $t > 1$:

- P moves in the negative direction when $1 < t < \frac{5}{3}$.
- P moves in the positive direction when $t > \frac{5}{3}$.
- Q moves in the negative direction when $1 < t < 2$.
- Q moves in the positive direction when $t > 2$.

So both particles move in the same direction when both velocities have the same sign.

FINAL ANSWER

$1 < t < \dfrac{5}{3}$ or $t > 2$

PAST PAPER QUESTION Q#: 23 Marks: 6

TOPIC: **Differentiation**

Jan 2022 P1HR - Jan 2022 | P1HR

23 Two particles, P and Q, move along a straight line.
The fixed point O lies on this line.

The displacement of P from O at time t seconds is s metres, where

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The displacement of Q from O at time t seconds is x metres, where

$$x = t^2 - 4t + 4 \quad \text{for } t > 1$$

Find the range of values of t where $t > 1$ for which both particles are moving in the same direction along the straight line.

(Total for Question 23 is 6 marks)

PAST PAPER SOLUTION Q#: 23 Marks: 6

TOPIC: **Differentiation**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Corrected from Functions to Differentiation.

Method

Particle P:

$$s = t^3 - 4t^2 + 5t$$

$$\frac{ds}{dt} = 3t^2 - 8t + 5$$

$$\frac{ds}{dt} = (3t-5)(t-1)$$

Particle Q:

$$x = t^2 - 4t + 4$$

$$\frac{dx}{dt} = 2t - 4 = 2(t-2)$$

For $t > 1$:

- P moves in the negative direction when $1 < t < \frac{5}{3}$.
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- Q moves in the negative direction when $1 < t < 2$.
- Q moves in the positive direction when $t > 2$.

So both particles move in the same direction when both velocities have the same sign.

FINAL ANSWER

$1 < t < \dfrac{5}{3}$ or $t > 2$

PAST PAPER QUESTION Q#: 24 Marks: 5

TOPIC: **Vectors**

Jan 2022 P1HR - Jan 2022 | P1HR

24 *OXYZ* is a trapezium.

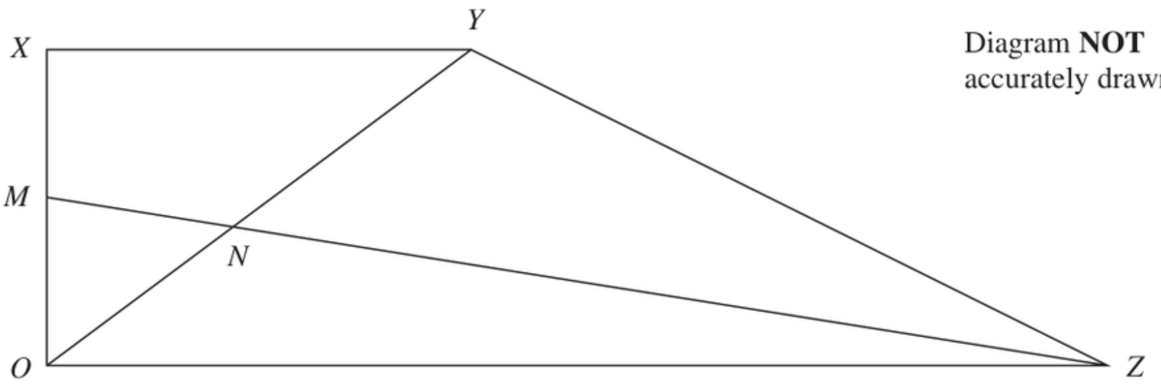

Diagram **NOT** accurately drawn

$\overrightarrow{OX} = \mathbf{a}$

$\overrightarrow{XY} = \mathbf{b}$

$\overrightarrow{OZ} = 3\mathbf{b}$

M is the midpoint of *OX*
N is the point such that *MNZ* and *ONY* are straight lines.

Given that $ON : OY = \lambda : 1$

use a vector method to find the value of λ

$\lambda = \ldots\ldots\ldots\ldots\ldots\ldots$

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION Q#: 24 Marks: 5

TOPIC: **Vectors**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION

Dr. Eslam Ahmed

Topic check. Moved from Coordinate Geometry to Vectors.

Method

Take O as the origin.

$$\vec{OX} = \mathbf{a}, \qquad \vec{XY} = \mathbf{b}, \qquad \vec{OZ} = 3\mathbf{b}$$

So

$$\vec{OY} = \mathbf{a} + \mathbf{b}$$

Since M is the midpoint of OX,

$$\vec{OM} = \frac{1}{2}\mathbf{a}$$

Given $ON : OY = \lambda : 1$,

$$\vec{ON} = \lambda(\mathbf{a} + \mathbf{b})$$

Point N also lies on MZ, so

$$\vec{ON} = \frac{1}{2}\mathbf{a} + t\left(3\mathbf{b} - \frac{1}{2}\mathbf{a}\right)$$

$$\vec{ON} = \left(\frac{1}{2} - \frac{t}{2}\right)\mathbf{a} + 3t\mathbf{b}$$

Equate coefficients with $\lambda\mathbf{a} + \lambda\mathbf{b}$:

$$\lambda = \frac{1}{2} - \frac{t}{2}$$

$$\lambda = 3t$$

So

$$t = \frac{\lambda}{3}$$

Substitute:

$$\lambda = \frac{1}{2} - \frac{1}{2}\left(\frac{\lambda}{3}\right)$$

$$\lambda = \frac{1}{2} - \frac{\lambda}{6}$$

$$\frac{7}{6}\lambda = \frac{1}{2}$$

$$\lambda = \frac{3}{7}$$

FINAL ANSWER

$\lambda = \frac{3}{7}$

PAST PAPER QUESTION Q#: 24 Marks: 5

TOPIC: **Vectors**

Jan 2022 P1HR - Jan 2022 | P1HR

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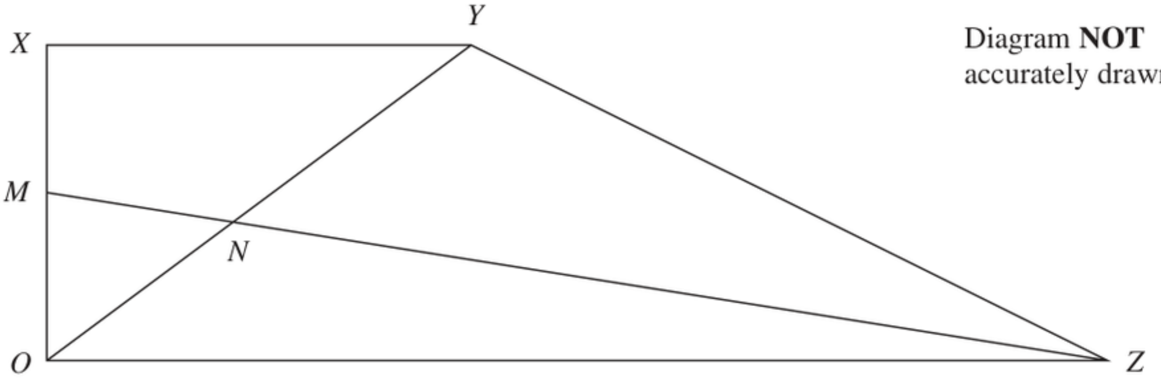

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$\lambda =$ ....

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION Q#: 24 Marks: 5

TOPIC: **Vectors**

Jan 2022 P1HR - Jan 2022 | P1HR

WORKED SOLUTION

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Point N also lies on MZ, so

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$$\vec{ON} = \left(\frac{1}{2} - \frac{t}{2}\right)\mathbf{a} + 3t\mathbf{b}$$

Equate coefficients with $\lambda\mathbf{a} + \lambda\mathbf{b}$:

$$\lambda = \frac{1}{2} - \frac{t}{2}$$

$$\lambda = 3t$$

So

$$t = \frac{\lambda}{3}$$

Substitute:

$$\lambda = \frac{1}{2} - \frac{1}{2}\left(\frac{\lambda}{3}\right)$$

$$\lambda = \frac{1}{2} - \frac{\lambda}{6}$$

$$\frac{7}{6}\lambda = \frac{1}{2}$$

$$\lambda = \frac{3}{7}$$

FINAL ANSWER

$\lambda = \frac{3}{7}$